

Lab 7: Comparison of Naive Bayes and Logistic Regression

We discussed naive Bayes and Logistic Regression classifiers in class. Naive Bayes is a Generative classifier, while Logistic Regression is a Discriminative classifier. In this lab, we will compare the performance of these two classifiers on [the 20Newsgroup dataset](#).

Preprocess the dataset by lowercasing and removing [stop words](#).

Create a hierarchy of classes as mentioned below:

comp

- comp.graphics
- comp.os.ms-windows.misc
- comp.sys.ibm.pc.hardware
- comp.sys.mac.hardware
- comp.windows.x

rec

- rec.autos
- rec.motorcycles
- rec.sport.baseball
- rec.sport.hockey

sci

- sci.crypt
- sci.electronics
- sci.med
- sci.space

politics

- talk.politics.misc
- talk.politics.guns
- talk.politics.mideast

religion

- talk.religion.misc
- alt.atheism
- soc.religion.christian

Tasks:

1. Naive Bayes Classification (3 Marks)

A. Train a naive Bayes classifier for each of the following tasks:

(You may use your earlier implementation)

(i) sci.med vs sci.space

(ii) comp.sys.ibm.pc.hardware vs comp.sys.mac.hardware

(iii) politics vs religion

(iv) politics vs sci

(v) politics vs rec vs religion vs sci

B. Evaluate the classifiers on the respective test datasets and report precision, recall, F1 score, along with the confusion matrix.

C. Interpret the classifiers and explain in your own words the behavior of these classifiers

2. Logistic regression (3 Marks)

A. Represent each document as a [TF-IDF vector](#). You are supposed to write your own TF-IDF Vectorizer.

B. Train Logistic Regression classifiers on the tasks mentioned in Step-1 using TF-IDF-based representation of documents

C. Evaluate the classifiers on the respective test datasets and report precision, recall, F1 score, along with the confusion matrix.

D. Interpret the classifiers and explain in your own words the behavior of these classifiers

3. Compare the performance of naive Bayes and Logistic Regression. Considering their underlying assumptions, explain the behavior of the classifiers across the tasks (4 Marks)

Please note that the tasks are purposefully vague. The lab will be evaluated based on the quality of your analysis.